Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State that you are treating air resistance as small enough to ignore.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your answer. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Warm-up: bring back an earlier idea

A **3 kg** test mass feels **15 N** toward a source. **Find g** and the force on a **7 kg** mass at the same point.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: a_x = **0**; a_y = -g; v_x = constant.
Upward positive; g = **10 m/s²**. Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at **3 m/s** and falls for **2 s**. Horizontal travel is **3×2=6 m**. With g=**10 m/s²** and no initial vertical velocity, its final vertical velocity is **20 m/s** down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after **2 s**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

One ball is dropped and another launched horizontally from the same height at the same instant. **Compare fall times** and explain.

Sentence starter: Both ... . They differ in ... because ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball rolls horizontally off a platform at **3 m/s** and is in the air for **2 s**. **Label the horizontal** and vertical motions. **Find the horizontal travel**. **Predict what changes** if the horizontal launch speed doubles while the platform height stays the same. Ignore air resistance.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a larger horizontal speed holds the ball up longer. **Explain which force or acceleration** would be needed for that statement.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A student says a **4 kg** cart under **24 N** net force accelerates at **96 m/s²**. **Name the error**, correct it and **name a check you can** now do independently. **Draw the forces** on the chosen object and **label their sources**.

Sentence starter: The step I would change is … . I would change it because I found … .

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two horizontal launches from the same height have speeds **2** and **6 m/s**. **Compare fall times** and ranges. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: Both … . They differ in … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two balls launch horizontally from the same platform at **2** and **5 m/s**. Their shared fall time is **3 s**. **Draw paths** and **compare ranges**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3